

A Tale of Two Riders: Cyclistic Bike-Share Analysis

Uncovering Urban Mobility Patterns through Data Science & Digital Humanities



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Focus: Public Policy, Digital Humanities, and
Creative Data Storytelling

Project Overview & Philosophy

More Than Just Sales: Viewing the City as an Archive

The Objective: This case study analyzes historical data from Cyclistic (Chicago) not just to increase sales, but to understand the deep behavioral divergence between distinct user groups.

The Unique Approach:

- Approaching the dataset through the lens of History and Public Policy.
- Treating bike data as an 'archive of city life.'
- Using inferential statistics to prove these groups inhabit the city in fundamentally different ways.



The Business Task

The Challenge: Converting Casual Riders into Members

The Stakeholder Goal: The Director of Marketing needs to maximize annual memberships for future company growth.

The Current Landscape: 'Casual' riders (leisure/tourists) are aware of the program but haven't committed to subscriptions.

The Central Question: How do 'Annual Members' and 'Casual Riders' use Cyclistic bikes differently?



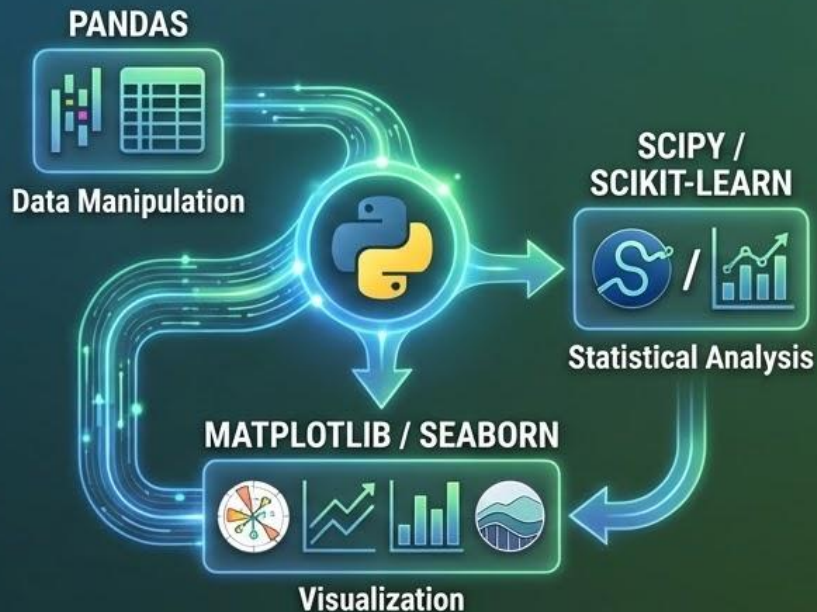
Methodology & Tech Stack

End-to-End Data Handling with Python

Rationale: Moving beyond spreadsheet limitations to handle millions of rows efficiently.

The Stack:

- **Data Manipulation:** Pandas (Merging 12 months of data, datetime conversion).
- **Statistical Analysis:** Scipy / Scikit-Learn (Validating behavioral differences).
- **Visualization:** Matplotlib & Seaborn (Statistical plots).



Data Source & Preparation Pipeline

From Raw Files to Reliable Features

Source: 12 months of previous trip data (Public data via Motivate International Inc.).

The Process:

- **Ingestion:** Iterated through and merged 12 separate CSV files into one massive dataframe.
- **Feature Engineering:** Created ride_length (ended_at - started_at) and extracted temporal features (day_of_week, month).
- **Sanity Check & Cleaning:** Removed 'administrative' rides (negative durations or <60s) and filtered missing geospatial coordinates.



Insight 1 - The “Pulse” of the City (Temporal)

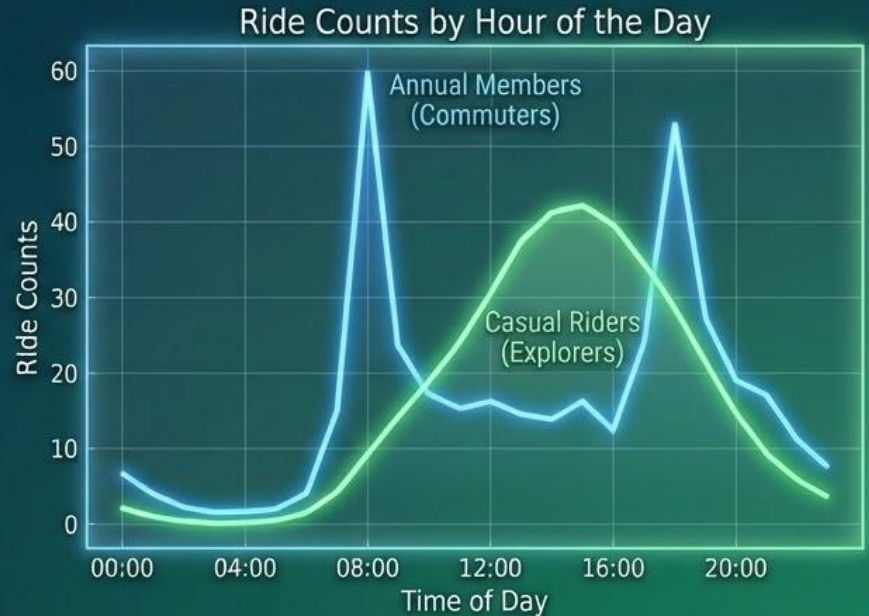
Two Different Schedules: Commuters vs. Explorers

Annual Members ("The Commuters"):

- Usage is highly predictable and efficient.
- Peaks: 8:00 AM and 5:00 PM on weekdays.

Casual Riders ("The Explorers"):

- Usage is driven by leisure and discovery.
- Peaks: Ramps up slowly daily; highest activity on weekends and sunny afternoons.



Insight 2 - Duration vs. Distance

Valuing Experience Over Efficiency

- **The Finding:** Casual riders take significantly longer rides than Members, even though they often cover less 'efficient' geographical distances.
- **Statistical Validation:** A T-test confirmed the statistical significance of these duration differences.
- **Interpretation:** Casuals value the experience of the ride itself, whereas Members value the destination.



Insight 3 - Station Geography

Mapping Motivation: The Leisure Corridor

Members concentrate near:

- Transit Hubs
- Commercial/Business Districts

Casuals cluster around:

- Parks
- Museums
- The Waterfront (The “Leisure Corridor”)

Members (Commuters)



Casuals (Explorers)



Humanizing the Insights through Digital Humanities

To bridge the gap between raw data and stakeholder understanding, creative deliverables were developed:

- **“The Tale of Two Riders” Infographic:** A visual persona guide using cartoons to contrast pain points and motivations of the “Commuter” vs. the “Explorer.”
- **The City Flow Map:** A visualization illustrating how the city “breathes” differently on Member-dominant weekdays vs. Casual-dominant weekends.



Strategic Recommendations

Meeting Casual Riders Where They Are

Based on behavioral divergence, a blanket approach will not work. We propose:



- **The “Weekend Warrior” Pass:** Create a Fri-Sun tier. Data shows Casuals are active then but feel an annual pass is “wasted” on workdays.



- **Gamified “History Hunter” Routes:** Leverage the desire for exploration via app-based curated routes to landmarks, incentivizing longer rides.



- **Strategic Digital Ads:** Shift ad spend to Friday afternoons/Saturday mornings, geofencing areas around parks and museums.



Thank You
Cyclistic